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Integrating large language models for enhanced predictive analytics in healthcare



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Physicians frequently confront time-sensitive decisions under uncertain conditions, necessitating reliable tools for forecasting clinical outcomes. Although clinical predictive models have the potential to assist in these critical decisions, their widespread adoption is hindered by complexities in data handling, model development, and integration into clinical workflows. This study introduces a novel framework (Hopkins LLM) leveraging structured electronic health records (EHRs) data to develop and deploy clinical large language models (LLMs) that act as multi-task-capable predictive engines to support clinically constrained decision-support tasks with minimal barriers to implementation. Employing the advanced LLaMA architecture, consisting of 7 billion parameters, our model was pre-trained on a comprehensive corpus and subsequently fine-tuned and tested on a dataset of 42,160 patients within Johns Hopkins Health System, addressing a spectrum of clinical and operational prediction tasks. We validated our model across three diverse external health systems and four key prediction tasks involving 1,329 patients, including 30-day all-cause readmissions, 90-day all-cause mortality, 30-day intensive care unit (ICU) admissions, and treatment recommendations. The proposed Hopkins-LLM framework achieved a mean area under the receiver operating characteristic curve (ROC-AUC) of 0.84 [0.82, 0.88], yielding a significant 0.28 advancement over zero-shot baseline LLMs ($p < 0.05$). These findings underscore the promise of LLMs as unified, user-friendly clinical prediction systems, adept at reasoning across diverse data sources to enhance decision-making at the point of care.

Physicians make complex decisions every day, integrating large volumes of heterogeneous clinical information¹. Yet the data needed to support sound decision-making are dispersed across medical histories, laboratory values, imaging reports, and other documentation within structured electronic health records (EHRs)². This fragmentation contributes to frequent decision-making errors, with cognitive mistakes in primary care estimated to affect roughly 20% of encounters³ and to account for about 17% of all adverse events in clinical practice⁴. To address these challenges, clinicians

and health systems have long relied on traditional rule-based clinical decision support systems (CDSSs)^{5,6} and, more recently, machine learning models trained on structured EHRs features to guide risk estimation and management decisions^{7–9}. However, these systems depend heavily on structured numerical inputs, which require extensive preprocessing and complex integration workflows¹⁰. Transformer architectures, the foundation of modern large language models, have recently been applied directly to longitudinal EHR and clinical time-series data to model patient trajectories

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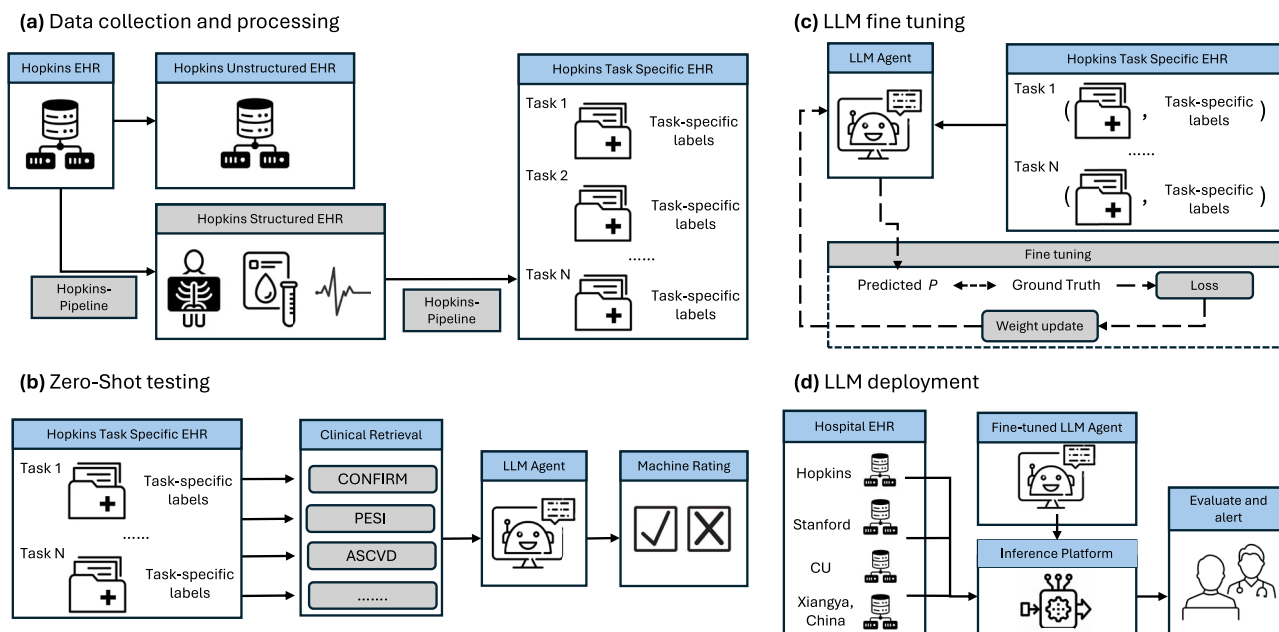


Fig. 1 | Schematic overview of the development and evaluation of Hopkins-LLM as a diagnostic generalist. a Data collection and processing: Unstructured EHRs from Johns Hopkins Health System were converted into structured inputs using Hopkins-Pipeline, and modality-specific datasets were constructed using imaging timestamps to align all kinds of clinical variables. **b** Zero-shot evaluation: Diagnostic testing was performed using either the full patient record or a clinically filtered subset identified through a clinical preference-aligned retrieval module. **c** Model fine-

tuning: The LLaMA-7B backbone was fine-tuned using chain-of-thought diagnostic rationales and aligned with hierarchical diagnostic and clinical preferences, using data from 42,160 imaging-triggered encounters. **d** Performance benchmarking: Hopkins-LLM was evaluated across internal (Johns Hopkins Health System) and external datasets from Stanford Hospital (Stanford), the UCHealth University of Colorado Hospital (CU), and Second Xiangya Hospital, China (Xiangya, China) under diverse distribution settings.

without manual feature engineering, improving predictive performance across multimodal records^{11–13}. However, their substantial computational demands, reliance on large datasets, and limited interpretability may constrain scalability, transparency, and real-time clinical deployment¹⁴. Consequently, despite thousands of predictive algorithms being developed and published, only a small fraction are ever deployed in real clinical settings, a persistent barrier often referred to as the “last-mile problem.”

Since the release of ChatGPT and other large generative language models in late 2022, natural language processing (NLP) has undergone a rapid transformation. Earlier pre-trained language models, such as BERT and BioBERT^{15,16}, introduced self-supervised learning on large corpora, but large-scale autoregressive models^{17,18} demonstrated unprecedented instruction following, zero-shot reasoning, and broad generalization. This shift has accelerated efforts to adapt LLMs for healthcare, leading to domain-specific systems such as ClinicalBERT¹⁹, NYUTron²⁰, GatorTron²¹, and BioGPT²². These models show early promise in simplifying task-specific pipelines and helping address the challenge in medical predictive analytics by enabling more flexible and generalizable clinical applications.

Despite rapid progress, large language model (LLM) integration into clinical workflows remains limited. Emerging federated learning frameworks for LLMs^{23–25} offer a potential pathway for collaborative model development across institutions while preserving data privacy and governance²⁶. Most existing studies focus on early-stage use cases, often involving ChatGPT^{17,27}, and few publicly available models are purpose-built for healthcare. Early efforts in medical question-answering and clinical dialogue^{28–30} show potential, but their effectiveness for clinical diagnostic reasoning and decision support is largely untested. Persistent risks of factual errors or “hallucinations”³¹ highlight the need for clinically aligned training that ensures safe and reliable instruction following³². Current evaluation methods, dominated by narrow automated benchmarks, underscore the need for broader, human-centered assessment of LLM performance in real clinical settings.

To address these challenges, we developed Hopkins-LLM (Fig. 1), a generalizable clinical language-model pipeline that integrates cohort

construction, multimodal data parsing, zero-shot inference, task-specific fine-tuning, and real-time internal and external deployment across four medical centers. The model is built on a 7-billion-parameter LLaMA backbone³³, was initially fine-tuned and evaluated on multimodal EHRs from 42,160 patients at Johns Hopkins Health System (JHHS), and was subsequently validated across three external health systems. Because clinical decision-making often begins when a patient undergoes a diagnostic test, we anchored cohort selection to common imaging encounters to create transparent and workflow-aligned entry points. Imaging provides a standardized moment when clinicians integrate history, laboratory data, and radiologic findings; selecting these encounters therefore yields a reproducible and clinically meaningful sampling frame. We focused on four high-volume modalities that span diverse organ systems and presentations: (1) CT pulmonary angiography (CTPA) for suspected pulmonary embolism (PE), (2) cardiac CT angiography (CCTA) for suspected coronary artery disease, (3) cardiac magnetic resonance imaging (Cardiac MRI) for evaluation of cardiomyopathy, structural abnormalities, or myocardial viability, and (4) right-upper-quadrant (RUQ) ultrasound for suspected acute cholecystitis.

From each imaging-triggered encounter, we aggregated a harmonized set of clinical inputs, including diagnostic reports, demographics, vital signs, laboratory results, ICD-10 codes, procedures, and longitudinal outcomes. This design provides a principled, disease-agnostic structure for evaluating model behavior while minimizing selection bias. To enable zero-shot diagnostic generalization, we introduced a unified clinical preference-alignment strategy combining (1) guidance from established diagnostic and prognostic criteria and (2) ICD-10 codes based hierarchical diagnostic preferences. Chain-of-thought (COT) fine-tuning³⁴ further encouraged reasoning patterns that align with clinical expectations³⁵. Ablation studies confirmed the additive contributions of each component.

We evaluated Hopkins-LLM using internal data from the Johns Hopkins Health System and external datasets from Stanford Hospital (Stanford), the UCHealth University of Colorado Hospital (CU), and

Second Xiangya Hospital, China (Xiangya). Across diagnostic and prognostic tasks, Hopkins-LLM outperformed traditional machine-learning and deep-learning baselines and maintained robust performance across institutions, modalities, and training regimes.

Results

Overview of the Hopkins-LLM framework and dataset characteristics

We introduce Hopkins-LLM, a clinical large language model designed as a diagnostic generalist capable of integrating heterogeneous clinical data and performing structured reasoning across multiple diagnostic contexts. The development pipeline includes four components: data collection and pre-processing, zero-shot evaluation, fine-tuning and alignment, and large-scale multi-institutional benchmarking (Fig. 1a–d).

In the first stage, we constructed a unified clinical corpus from real-world EHRs within the Johns Hopkins Health System. Using the Hopkins-Pipeline (Fig. 1a), unstructured notes, diagnostic reports, demographics, vital signs, laboratory results, ICD-10 codes, and imaging reports were standardized, temporally aligned, and converted into structured inputs. Imaging timestamps were used as anchors to synchronize clinical variables around each diagnostic study, producing patient-level structured records. We then defined several diagnostic scenarios triggered by the imaging exams patients received, each with corresponding expert-derived diagnostic or prognostic labels. This harmonized dataset served as the foundation for both zero-shot and fine-tuning experiments. The details of Hopkins-Pipeline are shown in the Supplementary Materials, Section 1.1 and Figure S1.

In the second stage (Fig. 1b), we evaluated Hopkins-LLM baselines' generalization and the impact of clinical preference alignment through a zero-shot protocol. Hopkins-LLM received either the full structured record or a filtered subset selected by a clinical preference-aligned retrieval module guided by established diagnostic severity scoring systems (for example, CONFIRM³⁶, PESI³⁷, ASCVD³⁸). For comparison, we included five additional LLMs: DeepSeek-R1³⁹, GPT-4.1¹⁷, Gemma-3⁴⁰, Med-Palm⁴¹, and Clinical Camel⁴². The details of the used diagnostic framework and LLMs are in the Methods Section and in Supplementary Material, Section 6, respectively. Outputs were scored automatically using machine-based metrics for diagnostic accuracy and consistency, enabling quantitative comparison between unconstrained reasoning and clinical domain-guided inference. This stage assessed the extent to which LLMs can perform clinically coherent reasoning without task-specific training.

In the third stage (Fig. 1c), Hopkins-LLM was fine-tuned on 42,160 imaging-triggered encounters using chain-of-thought supervision to promote explicit diagnostic rationales. A unified clinical preference-alignment strategy, combining guidance from widely adopted diagnostic criteria and hierarchical constraints from the ICD-10 ontology, was applied to achieve stable clinical alignment. These steps enhanced the model's interpretability, reliability, and adherence to clinically meaningful reasoning patterns.

Finally, in the deploying and benchmarking stage (Fig. 1d), Hopkins-LLM was evaluated across internal JHHS datasets and external cohorts, including 1329 patients who underwent CTPA for evaluation of PE from the UHealth University of Colorado Hospital, 65 patients who underwent cardiac MRI and 81 CCTA from Second Xiangya Hospital, and 108 patients who underwent RUQ ultrasound for cholecystitis evaluation from Stanford Hospital. Evaluation covered diverse institutional, demographic, and modality distributions. Hopkins-LLM consistently outperformed domain-specific machine-learning models, deep-learning baselines, and general-purpose LLMs across diagnostic accuracy, interpretability, and prognostic prediction. External validation studies further demonstrated that Hopkins-LLM can assist physicians as a generalist AI system across a range of diagnostic tasks.

Overall Performance on Four Predictive Tasks

To evaluate the breadth of Hopkins-LLM's clinical applicability, we assessed its performance across four predictive tasks, including 90-day mortality, 30-day ICU admission, 30-day hospital readmission, and treatment recommendation (Fig. 2). Each task was benchmarked under three configurations:

(1) **fine-tuned Hopkins-LLM**, (2) **zero-shot Hopkins-LLM**: LLM with structured EHRs inputs and using a preference-aligned retrieval module in a zero-shot setting, and (3) **zero-shot LLM**: LLM with unstructured EHRs and without using a preference-aligned retrieval module. All models used identical features and standardized evaluation metrics. LLaMA-7B is used as the baseline LLM. Across tasks, fine-tuned Hopkins-LLM achieved a mean AUC of 0.84 ± 0.03 , corresponding to an average 9.2% improvement over zero-shot Hopkins-LLM and an average 27.9% improvement over zero-shot LLM, demonstrating strong generalizability and clinically coherent reasoning.

Mortality prediction. Guided by ASCVD and PESI diagnostic severity scoring systems, Hopkins-LLM predicted in-hospital and 90-day post-discharge mortality with a median AUC of 0.84 ± 0.02 , a 7.9% improvement over zero-shot Hopkins-LLM and a 21.2% improvement over zero-shot LLM. For temporal survival estimation (<30, 30–90, >90 days), reframed as a multiclass task, the model achieved a median accuracy (ACC) of 0.73 (Fig. 2b), indicating well-calibrated temporal risk stratification. **ICU admission and readmission.** Hopkins-LLM predicted 30-day ICU admissions and all-cause 30-day readmissions using binary classification. For ICU admission, the model reached a median AUC of 0.86 ± 0.02 (a 10.7% improvement over zero-shot Hopkins-LLM). For 30-day readmission, it achieved a median AUC of 0.85 ± 0.02 (a 7.2% improvement over zero-shot Hopkins-LLM). **Treatment recommendation.** In the RUQ ultrasound cohort, Hopkins-LLM achieved a median AUC of 0.79 ± 0.01 , representing a 9.1% improvement over zero-shot Hopkins-LLM.

Performance on Internal Datasets across Diagnostic Tasks

To evaluate Hopkins-LLM within the development environment, we first assessed performance on internal datasets that maintained identical clinical variable formats across CTPA, CCTA, cardiac MRI, and RUQ ultrasound. Across all tasks (as shown in Fig. 3), fine-tuned Hopkins-LLM consistently outperformed both the structured zero-shot Hopkins-LLM and the unstructured zero-shot LLM settings. Evaluation metrics included AUC, ACC, and F1-score (0.5 threshold). Full results appear in Supplementary Materials, Table S5.

CTPA. For mortality prediction, fine-tuned Hopkins-LLM achieved an AUC of 0.85 (ACC 0.78; F1 0.78), outperforming the structured zero-shot Hopkins-LLM (AUC 0.80) and the unstructured zero-shot LLM (AUC 0.62). ICU admission prediction similarly favored fine-tuned Hopkins-LLM (AUC 0.85 vs. 0.79 and 0.60). For 30-day readmission, the fine-tuned model achieved an AUC of 0.86 (ACC 0.79; F1 0.79), again exceeding both zero-shot variants. **CCTA** For ICU admission, fine-tuned Hopkins-LLM achieved an AUC of 0.87 (ACC 0.77; F1 0.76), outperforming structured (AUC 0.77) and unstructured (AUC 0.55) models. Mortality prediction also improved (AUC 0.82 vs. 0.75 and 0.64). For 30-day readmission, the fine-tuned model achieved an AUC of 0.82 (F1 0.74), compared to 0.73 and 0.57 for the zero-shot Hopkins-LLM and zero-shot LLM. **Cardiac MRI.** On cardiac MRI, fine-tuned Hopkins-LLM reached an AUC of 0.83 (ACC 0.79; precision 0.80), outperforming the structured zero-shot Hopkins-LLM model (AUC 0.76) and zero-shot LLM (AUC 0.52), demonstrating strong robustness across imaging modalities. **Right upper quadrant (RUQ) Ultrasound.** For treatment-outcome prediction, Hopkins-LLM achieved an AUC of 0.79 (ACC 0.76; F1 0.75), surpassing both structured (AUC 0.69) and unstructured (AUC 0.53) zero-shot LLM.

Across eight internal prediction tasks spanning pulmonary, cardiac, and ultrasound cohorts, fine-tuned Hopkins-LLM improved AUC by 0.08–0.15 over the structured zero-shot Hopkins-LLM variant and by 0.20–0.30 over unstructured zero-shot LLM. These consistent gains demonstrate the value of task-specific fine-tuning and structured EHRs alignment in enhancing multimodal reasoning and predictive performance.

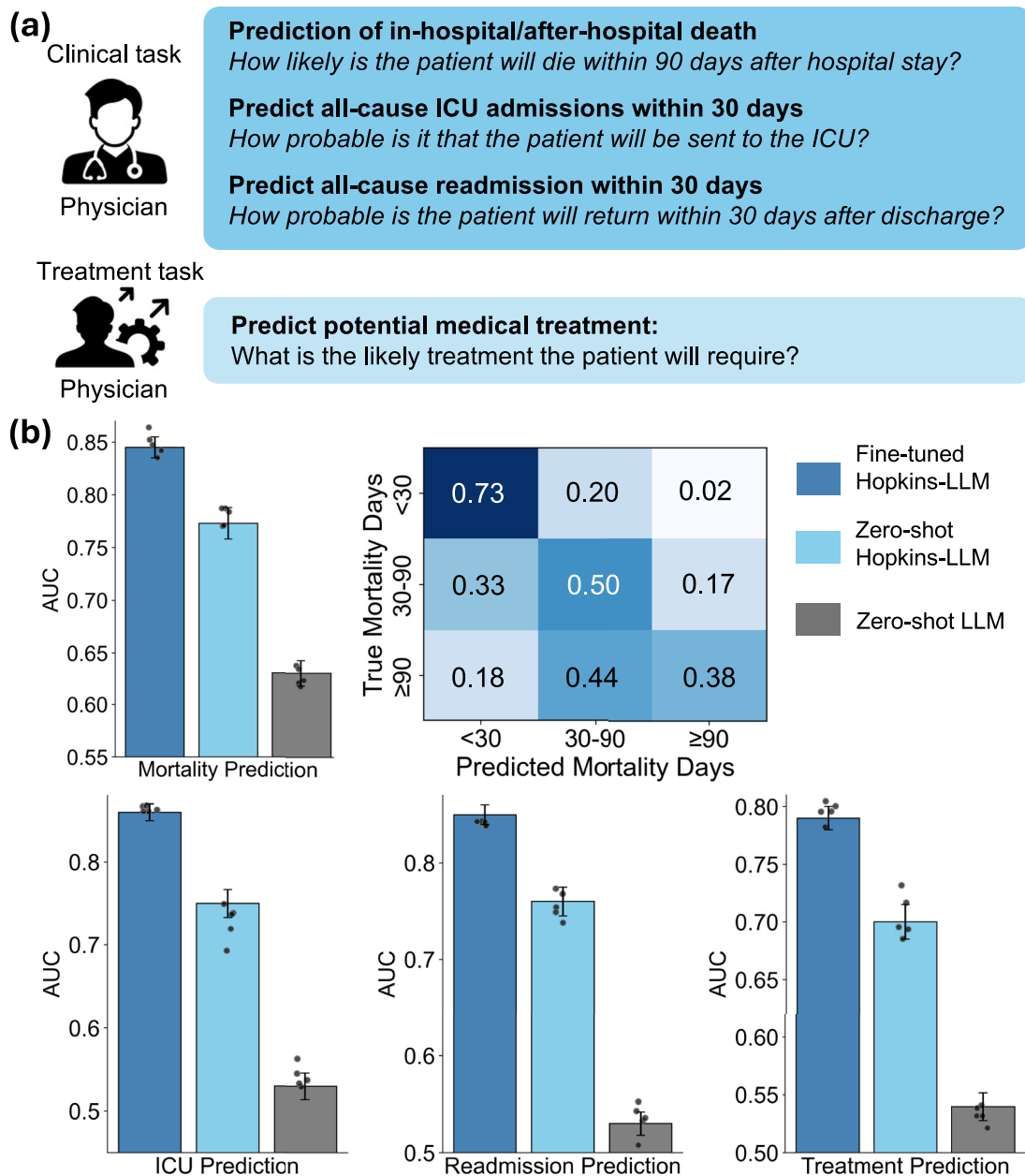


Fig. 2 | Overall diagnostic and prognostic performance of Hopkins-LLM. a Four predictive tasks were evaluated across real-world clinical outcomes: 90-day all-cause mortality prediction (in-hospital and post-discharge), 30-day readmission, 30-day ICU admission, and treatment recommendation. **b** Performance comparison with baseline models. The confusion matrix (center) depicts temporal calibration for

mortality across three outcome bins (<30, 30–90, and ≥90 days). For all tasks, the height of each bar represents the median AUC, and the half-width of the error bar indicates 1 s.d. The grey points denote individual data points from $n = 5$ runs conducted with distinct random seeds.

Performance on External Datasets across Diagnostic Tasks

To evaluate the breadth and robustness of Hopkins-LLM across diverse clinical contexts, we assessed its performance on prediction tasks spanning external datasets. We first compare six large language models (LLaMA, GPT, DeepSeek, Gemma, Med-Palm and Clinical Camel) on external datasets in both zero-shot Hopkins-LLM setting (with clinical guidance alignment settings) and the zero-shot LLM setting (with unstructured input), Fig. 1b. The detailed results for each LLM model are summarized in the Supplementary Materials, Table S6 to Table S9.

Across internal datasets, zero-shot Hopkins-LLM with a clinical preference-aligned retrieval module consistently improved discriminative performance (Fig. 4a). Averaged across eight internal tasks, LLaMA increased from 0.57 ± 0.08 to 0.75 ± 0.05 , GPT from 0.69 ± 0.06 to

0.77 ± 0.05 , DeepSeek from 0.66 ± 0.07 to 0.76 ± 0.05 , Gemma from 0.63 ± 0.08 to 0.72 ± 0.05 , Med-PaLM from 0.73 ± 0.03 to 0.79 ± 0.02 , and Clinical Camel from 0.71 ± 0.04 to 0.76 ± 0.03 . In addition to higher mean performance, preference-aligned retrieval reduced typical task-level uncertainty: mean per-task error decreased from 0.076 to 0.046 for LLaMA, 0.060 to 0.053 for GPT, 0.068 to 0.049 for DeepSeek, 0.078 to 0.051 for Gemma, 0.034 to 0.025 for Med-PaLM and 0.043 to 0.027 for Clinical Camel. External validation mirrored these trends, indicating that clinically aligned retrieval generalizes across institutions and imaging modalities and highlighting the robustness of the Hopkins-LLM framework in adapting to heterogeneous data distributions. Collectively, these results show that structured preference alignment narrows the gap between zero-shot LLM inference and domain-optimized clinical reasoning.

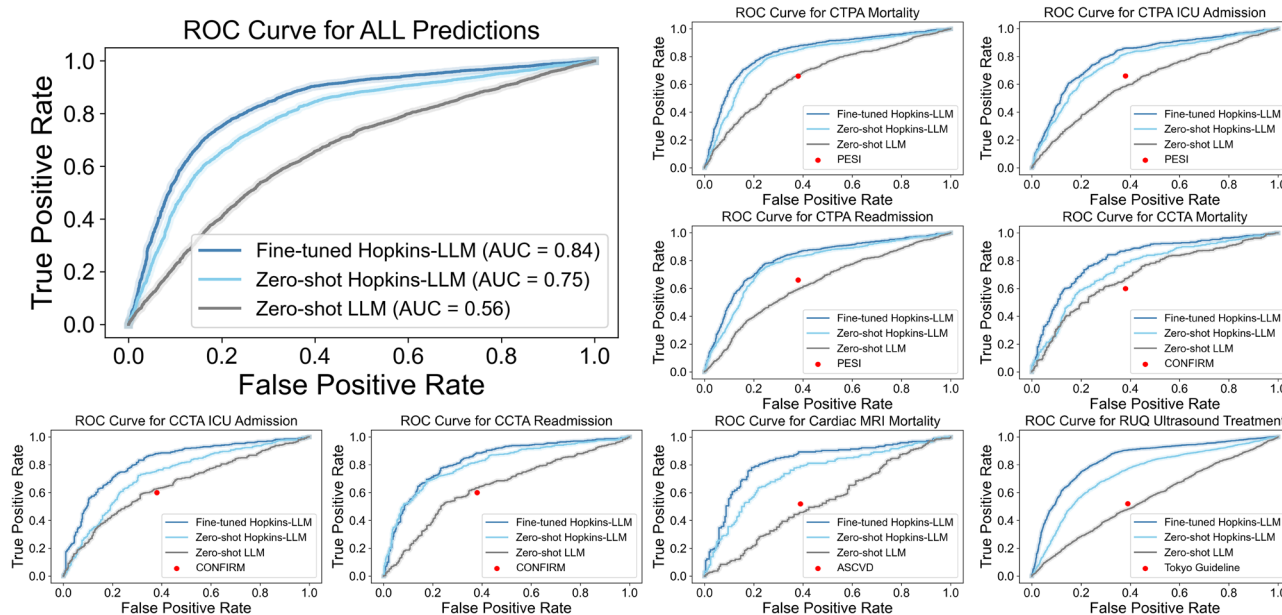


Fig. 3 | ROC curves for internal validation across diagnostic tasks. Fine-tuned Hopkins-LLM consistently outperformed zero-shot Hopkins-LLM and zero-shot LLM across all prediction tasks. The model achieved higher AUCs for CTPA (mortality, ICU admission, readmission), CCTA and cardiac MRI (mortality, ICU

admission, readmission), and RUQ ultrasound-based treatment prediction, surpassing standard clinical risk scores (PESI, CONFIRM, ASCVD, Tokyo Guidelines). Pooled analysis (left) shows overall discrimination (AUC = 0.84 vs. 0.75 and 0.56).

As shown in Fig. 4b, we assessed the fine-tuned Hopkins-LLM for cross-institutional generalization by comparing internal versus external test performance on diagnostic tasks. Across all tasks, the model maintained strong discriminative ability externally, with only modest declines in mean ROC-AUC and no statistically significant differences (all comparisons, ns). On internal datasets, the fine-tuned model achieved AUCs of 0.85 ± 0.07 , 0.85 ± 0.04 , 0.86 ± 0.03 , 0.83 ± 0.04 , 0.87 ± 0.03 , 0.82 ± 0.03 , 0.84 ± 0.04 , and 0.79 ± 0.05 across the eight prediction tasks, respectively. When evaluated on external cohorts, AUCs remained high at 0.83 ± 0.05 , 0.84 ± 0.05 , 0.87 ± 0.05 , 0.82 ± 0.03 , 0.85 ± 0.08 , 0.81 ± 0.06 , 0.82 ± 0.04 , and 0.77 ± 0.09 . Paired internal-external differences were modest (mean Δ AUC = 0.013), indicating that the fine-tuned Hopkins-LLM preserves predictive fidelity across unseen clinical environments and is suitable for cross-institutional deployment. The detailed results, including AUC, area under the precision-recall curve (AUPRC), ACC, and F1, for each LLM model are summarized in the Supplementary Materials, Table S12.

Multimodal Input Contribution Across Different LLMs

We evaluated how the progressive addition of clinical information influences diagnostic accuracy across four LLMs, including LLaMA, GPT, DeepSeek, Gemma, Med-Palm and Clinical Camel, across clinical contexts defined by the diagnostic imaging studies patients underwent. Each model was tested under hierarchical input settings that began with patient information alone, followed by the sequential inclusion of laboratory data, diagnosis ICD-10 codes, and imaging reports.

Across all models (Fig. 5), diagnostic accuracy improved steadily as more data sources were incorporated, underscoring the importance of multimodal integration for clinical reasoning. For LLaMA, accuracy increased from 50.2% with patient demographic information alone to 71.5% with imaging reports added, reaching 75.7% with full multimodal input. GPT showed similar gains, rising from 51.3% to 70.4% and ultimately 78.7%. DeepSeek improved from 48.4% to 66.6% with stepwise input expansion and reached 73.6% with complete multimodal information. Gemma increased from 50.0% to 67.1%, peaking at 77.7% with all information combined. Med-Palm increased from 52.6% to 67.0%, peaking at 78.1% with all information combined. Clinical Camel increased from 48.7% to 65.8%, peaking at 76.9% with all information combined. Overall, these

results demonstrate that integrating structured data (laboratories, diagnosis codes) with unstructured imaging and narrative information substantially enhances diagnostic accuracy across different LLM architectures.

Comparison with Traditional and Deep Learning Models

We compared Hopkins-LLM with conventional machine-learning models (Logistic Regression (LogReg), Random Forest (RF), XGBoost, SVM, MLP) and transformer-based deep learning baselines across four imaging-triggered clinical contexts defined by the study indication (CTPA, CCTA, cardiac MRI, RUQ ultrasound). Performance was summarized as the change in diagnostic accuracy (%) of each comparator model relative to the fine-tuned Hopkins-LLM; negative values therefore denote higher accuracy for Hopkins-LLM, and positive values indicate the comparator performed better (Fig. 6). The detailed results of ACC and AUC for each LLM model are summarized in the Supplementary Materials, Table S13 and Table S14.

On internal datasets, traditional models showed a modest advantage when only diagnostic information (DI) was available (mean difference $+1.51 \pm 0.94\%$, $p < 0.05$). Adding laboratory data eliminated this advantage (mean $-0.23 \pm 1.10\%$, no statistical significance), and incorporating physician examination (ICD-10 codes) findings shifted the balance in favor of Hopkins-LLM (mean $-1.82 \pm 1.09\%$, $p < 0.01$). With full multimodal input (All Info), Hopkins-LLM demonstrated a marked benefit (mean $-7.93 \pm 2.11\%$, $p < 0.001$), indicating substantially higher diagnostic accuracy than all baselines. External validation mirrored these trends. Comparator models retained a small edge in DI-only settings (mean $+2.66 \pm 0.82\%$, $p < 0.05$) and a slight advantage with laboratory data (mean $+0.54 \pm 1.01\%$, no statistical significance), but Hopkins-LLM surpassed them once physician examination findings were included (mean $-1.59 \pm 1.23\%$, $p < 0.01$). With full multimodal input, Hopkins-LLM achieved a large performance gain (mean $-6.03 \pm 2.15\%$, $p < 0.001$).

Across internal and external evaluations, Hopkins-LLM consistently improved with increasing clinical context, outperforming LogReg, RF, XGBoost, SVM, MLP, and transformer baselines in multimodal settings. These results highlight the model's strength in integrating heterogeneous clinical information and its robustness across indications and domains.

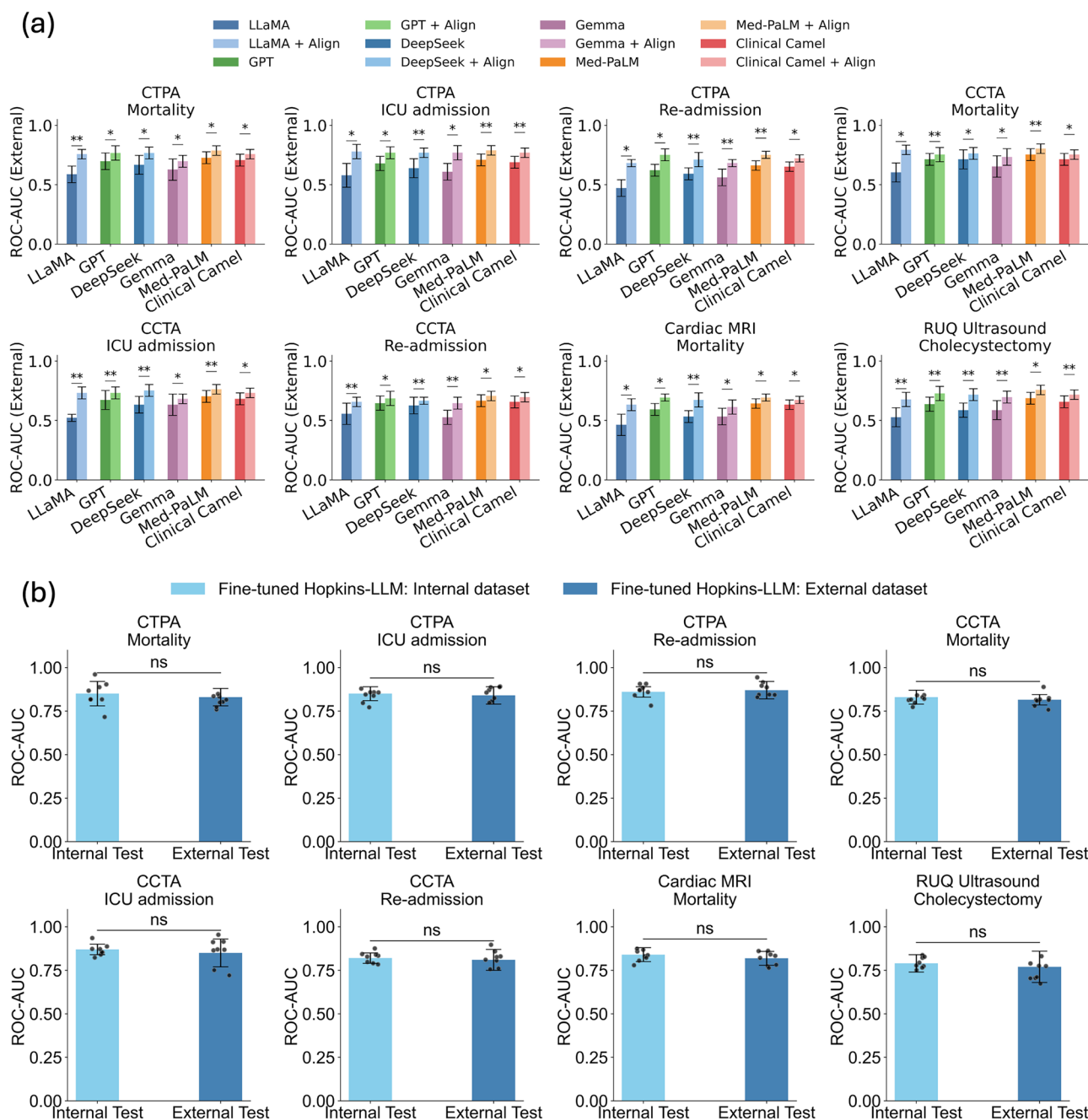


Fig. 4 | External validation of Hopkins-LLM for multimodal prognostic prediction. **a** Comparison of zero-shot Hopkins-LLM (with the clinical preference-aligned retrieval module) and baseline LLMs for prognostic prediction across five multimodal clinical tasks. External test performance for four LLM models, LLaMA, GPT, DeepSeek, and Gemma, before and after domain-specific alignment (+ Align).

Alignment consistently improved prognostic accuracy across all eight outcome tasks. **b** Internal vs. external ROC-AUC comparison for fine-tuned Hopkins-LLM across eight outcome tasks, demonstrating strong generalization with modest performance differences (ns: not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

Discussion

We present Hopkins-LLM, a clinically aligned large language model designed to perform generalizable diagnostic and prognostic reasoning across diverse medical settings. Evaluated on 90-day mortality, 30-day readmission, 30-day ICU admission, and treatment-recommendation tasks, and tested across multiple imaging-anchored cohorts, Hopkins-LLM consistently outperformed traditional machine-learning models, deep-learning baselines, and unaligned LLMs on both internal and external datasets. Comparative analyses using four different LLM architectures, evaluated with and without our clinical preference-alignment strategy, further demonstrated how structured clinical priors enhance diagnostic coherence. Fine-tuned Hopkins-LLM achieved the strongest overall performance,

supporting its potential as a generalist AI system capable of integration into real-world clinical workflows.

Accurate diagnosis remains central to medical practice but is vulnerable to human error, time pressure, and information overload, often contributing to delayed or incorrect treatment. Prior clinical decision-support systems, spanning rule-based CDSSs^{5,6} to machine-learning models trained on structured EHRs features^{43,44}, have shown promise yet remain constrained by narrow task domains, limited adaptability, and poor scalability across specialties. The emergence of medical LLMs^{19–22} has expanded the possibilities for broad clinical applicability, enabling tasks such as conversational assistance, summarization, and interactive decision support⁴⁵. However, general-purpose LLMs continue to struggle with clinical depth,

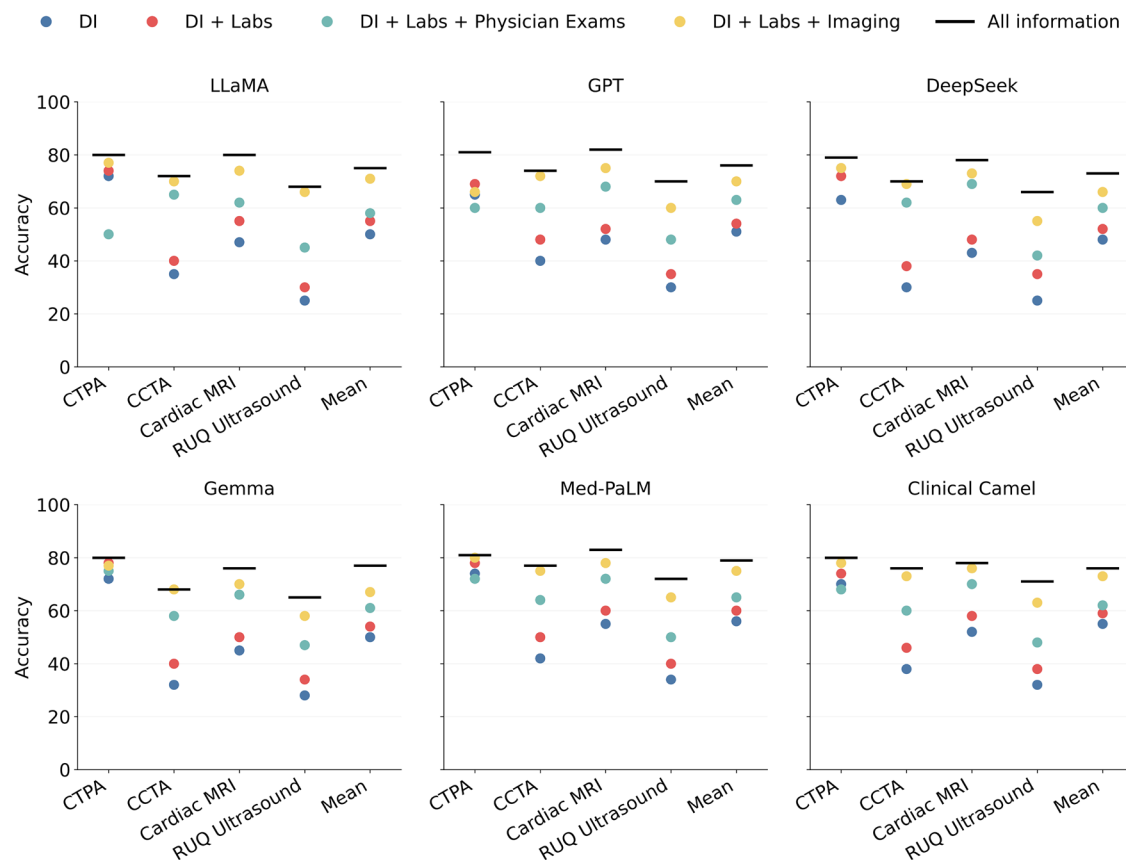


Fig. 5 | Diagnostic accuracy across multimodal input settings for four LLMs (LLaMA, GPT, DeepSeek, Gemma, Med-PaLM and Clinical Camel). Each model was evaluated with progressively enriched clinical information: demographic

information (DI), DI + laboratory data (Labs), DI + diagnosis codes (ICD-10) from physician exams, DI + imaging reports, and full multimodal context ("All information"). Accuracy increased consistently with richer inputs.

often producing factual inaccuracies, incomplete reasoning, or inconsistent conclusions^{46–48}. Hopkins-LLM seeks to address these limitations by combining structured clinical guidance, preference-alignment mechanisms, and targeted fine-tuning to enhance factual reliability and improve reasoning across varied diagnostic contexts.

A key motivation for adopting an LLM-based framework is not solely to maximize predictive performance on individual clinical outcomes, but to support generalizable, multi-task clinical decision support operating directly on heterogeneous clinical text. While simpler models such as logistic regression, random forests, or gradient-boosted trees can achieve competitive performance for well-defined prediction tasks using curated structured features⁴⁹, they are inherently limited in their ability to integrate nuanced clinical context, longitudinal narrative information, and cross-task reasoning without extensive manual feature engineering⁵⁰. In contrast, LLMs, including our Hopkins-LLM, learn rich representations directly from raw clinical text, enabling a single LLM backbone to flexibly support multiple clinical-relevant tasks through lightweight fine-tuning or task adaptation⁵¹. This design reduces the need for task-specific pipelines and facilitates more efficient adaptation to new clinical questions or external institutions. Importantly, our goal is not to replace traditional ML models, which remains highly effective for targeted risk prediction, but to complement them by addressing clinical scenarios that require synthesis of complex, narrative, and temporally evolving information. By framing the LLM as a clinician-facing decision-support tool rather than an autonomous decision-maker, we emphasize its role in augmenting clinical workflows, consistent with prior work on safe and effective clinical deployment of clinical AI systems.³²

A major strength of Hopkins-LLM is its use of structured clinical information throughout development. Unlike generic generative models,

which depend heavily on prompting and may produce unstable outputs^{53,54}, Hopkins-LLM integrates clinical priors, multimodal EHRs inputs, and hierarchical diagnostic preferences directly into the inference pipeline. This design enables rapid prototyping of new clinical tasks, facilitates integration with existing informatics systems, and helps overcome the gap between model development and real-world clinical deployment. To promote safety in real-world use, continuous supervision and monitoring are required to track human-AI interactions and mitigate model drift over time; details of this monitoring approach are provided in Supplementary Information.

Our findings also demonstrate the value of high-quality, domain-relevant data and focused fine-tuning. Although Hopkins-LLM (7B parameters) is substantially smaller than current large-scale (>70B) models, it achieved strong diagnostic performance with far lower computational demands. Each fine-tuning run required approximately 25 h on NVIDIA A100 GPUs, substantial but markedly lower than the requirements for most industry-scale LLMs. These results reinforce the principle that high-quality, in-domain data can be more impactful than sheer model size. For institutions with limited computational resources, we recommend fine-tuning a pretrained model—preferably one initialized on clinical text, though general models also perform strongly with sufficient in-domain adaptation. All training details are provided in Supplementary Information.

Hopkins-LLM was deliberately designed to prioritize clinical interpretability and accountability by operating on structured inputs derived from established disease-specific guidelines and severity scoring systems, such as the PESI³⁷ for CT pulmonary embolism. These inputs, including laboratory measurements, diagnosis codes, and imaging-derived summaries, allow model predictions to be traced to clinically recognizable variables, in contrast to end-to-end deep learning approaches that operate on unfiltered data representations. The model is further constrained to clearly

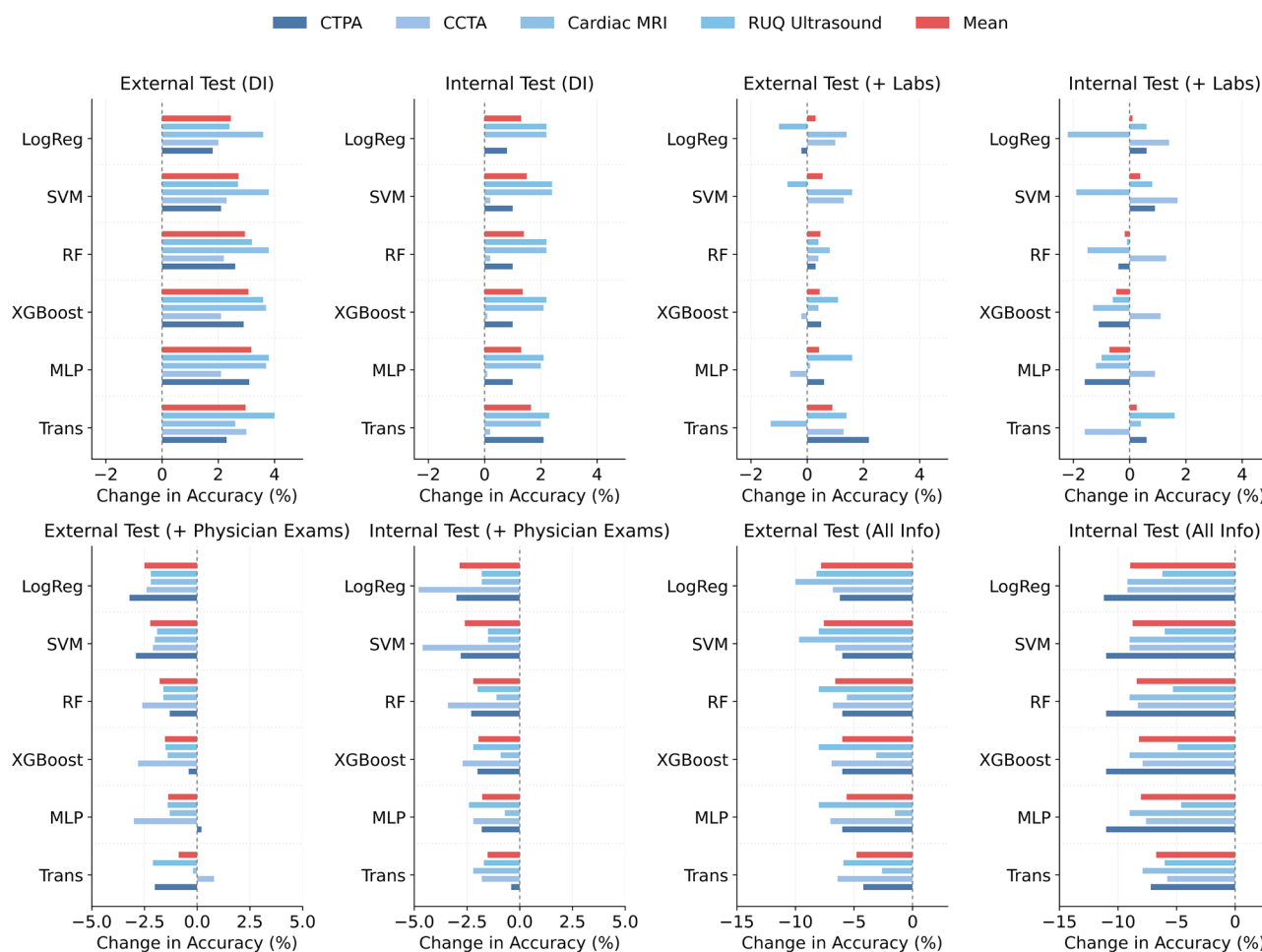


Fig. 6 | Comparison of Hopkins-LLM with traditional machine-learning and deep-learning baselines across multimodal prognostic tasks. Bars represent changes in diagnostic accuracy for internal and external evaluations under varying auxiliary input settings (demographic information (DI), laboratory data (Labs), diagnosis ICD-10 codes (from Physician Exams), and full multimodal context (including imaging reports)). Across all modalities, including CT pulmonary

embolism, cardiac CT angiography, cardiac MRI, and RUQ ultrasound, fine-tuned Hopkins-LLM consistently outperformed logistic Regression (LogReg), Random Forest (RF), XGBoost, SVM, MLP, and transformer-based models ("Trans"), demonstrating superior multimodal integration and more robust generalization across institutions.

defined clinical prediction tasks, such as mortality and ICU admission, rather than open-ended recommendations, which reduces the risk of hallucination and enables systematic auditing against ground-truth outcomes and accepted clinical standards⁵⁵. In this constrained prediction setting, model failures primarily manifest as misclassification or uncertainty in borderline cases with conflicting or incomplete structured inputs, rather than hallucination in the form of fabricated clinical facts or unsupported recommendations. Although this study does not introduce a new post hoc explainability method, the structured nature of the framework readily supports established interpretability tools for tabular prediction models, and task-specific confidence scores provide an explicit measure of prediction uncertainty. In addition, the multimodal ablation analyses (Fig. 5) offer global interpretability by quantifying the incremental contribution of distinct clinical information sources. Importantly, Hopkins-LLM is intended as a clinical decision-support tool within a human-in-the-loop paradigm¹⁴, with clinicians retaining ultimate responsibility for decision-making; future work will therefore focus on calibration, uncertainty estimation, and clinician-facing explanation interfaces to support safe and responsible clinical translation.

Hopkins-LLM also exhibits flexibility across specialty-specific diagnostic contexts with minimal prompting, suggesting its potential role as a clinical co-pilot. By synthesizing multimodal clinical data, it can propose differential diagnoses, refine prognostic assessments, and support

physicians, particularly trainees or non-specialists, during complex clinical decision-making. While our results provide evidence of robust performance across four independent health systems, larger randomized studies and broader real-world evaluations will ultimately be required to quantify clinical and operational impact.

Moreover, recent evidence suggests that LLMs applied to clinical decision-making may exhibit systematic biases arising from training data, model design, and deployment context, potentially leading to differential recommendations across patient groups based on non-clinical attributes such as race, gender, or socioeconomic status⁵⁶. These biases may reinforce existing disparities or introduce inconsistent reasoning patterns that affect diagnostic and management decisions⁵⁷. Hopkins-LLM is designed to address these risks through structured multimodal integration, guideline-informed preference alignment, and imaging-anchored cohort construction, which ground model reasoning in clinically verifiable evidence rather than unconstrained generative inference. This design promotes more consistent, transparent, and institutionally robust decision support across diverse patient populations.

Despite its strengths, several limitations warrant discussion. First, our imaging-anchored cohort design, chosen to align with real-world diagnostic workflows, may introduce selection bias toward patients who undergo advanced imaging. These cohorts reflect clinical suspicion for pulmonary embolism, coronary artery disease, cholecystitis, and cardiac pathology, and

therefore may not capture broader populations without imaging. Second, cardiac MRI, with its heterogeneous indications, introduces inherent variability despite standardized data harmonization. Third, although our multimodal inputs include rich structured and unstructured EHRs data, they exclude potentially informative modalities such as ECG waveforms, raw imaging pixels, genomic data, and clinical dialogues, all of which may enhance downstream reasoning. Lastly, while Hopkins-LLM demonstrated strong cross-institutional generalization, additional validation across diverse populations and care settings is required before deployment at scale.

Future advances may further extend Hopkins-LLM through integration with medical vision-language models (VLMs), which have shown strong performance in radiology, pathology, and echocardiography^{58–61}. By leveraging LLM-driven clinical knowledge and reasoning, VLMs can support zero-shot visual interpretation, multimodal grounding, and image-to-text generation. Agent-based LLM systems^{62,63} and human-in-the-loop refinement strategies⁶⁴ also offer promising paths toward personalization, enabling models to adapt to physician preferences and continuously improve based on clinical feedback. Such capabilities will be central to deploying generalist clinical AI systems safely and effectively, maximizing their utility in real-world diagnostic support and medical education. Finally, although mortality, readmission, and treatment prediction are well-established outcome measures for evaluating clinical risk models, future work will extend this framework to more proximal and directly actionable endpoints, including therapy escalation, ICU transfer timing, and complication onset, as multimodal annotation quality and temporal labeling mature.

Methods

Hopkins-LLM is a generalizable clinical language-model framework that fine-tunes pre-trained large language models on structured and unstructured clinical data (Fig. 1). The approach is model-agnostic and requires no modification to the underlying LLM architecture, enabling seamless adaptation to both general-purpose and clinically oriented text models. We trained and evaluated Hopkins-LLM on diagnostic and prognostic prediction tasks using four independent, imaging-anchored cohorts drawn from four institutions: CT pulmonary angiography (CTPA), cardiac CT angiography (CCTA), cardiac MRI, and right-upper-quadrant (RUQ) ultrasound.

Ethics Statement

Ethics approval and consent to participate: This study was approved by the Johns Hopkins Institutional Review Board with reference numbers IRB00424745 (Radiology foundation model for chest and cardiac image interpretation) and IRB00284909 (Artificial Intelligence Based on Imaging for Abdominal Oncology). External datasets are under the Data Users Agreement with each University/Hospital.

Internal dataset

We constructed a comprehensive multimodal imaging-EHRs dataset within the Johns Hopkins Health System to evaluate LLM-based clinical reasoning, with and without the proposed clinical preference-aligned retrieval module. The dataset spans four imaging modalities: CTPA, CCTA, cardiac MRI, and RUQ ultrasound. This design enables unified evaluation across cardiovascular, pulmonary, and abdominal disease contexts.

- CTPA Dataset: The internal CTPA cohort includes 14,984 patients evaluated for suspected pulmonary embolism. Each case is paired with structured and unstructured EHRs elements, including radiology reports, initial clinical notes, echocardiography summaries, medication history, CPT procedures, and ICD-10 diagnoses. Outcome labels include mortality, ICU admission, and 30-day readmission.
- CCTA Dataset: The internal CCTA cohort includes 6,044 patients assessed for acute or subacute cardiovascular conditions. All encounters include synced EHRs data, radiology reports, laboratory results,

procedures, and diagnostic codes. Outcomes include mortality, ICU admission, and long-term heart disease.

- Cardiac MRI Dataset: The internal cardiac MRI cohort consists of 11,176 patients undergoing cardiac MRI for suspected or known cardiovascular disease. It includes the same multimodal EHRs linkages as CTPA (clinical notes, radiology reports, medications, CPT procedures, ICD-10 diagnoses). The primary outcome is mortality.
- RUQ Ultrasound Dataset: The internal RUQ ultrasound cohort includes 9,956 patients undergoing RUQ ultrasound for evaluation of suspected cholecystitis. Each case contains radiology reports, clinical metadata, and EHRs-linked information. The primary outcome is treatment decision (cholecystectomy vs. no intervention).

All internal datasets were obtained under Johns Hopkins Institutional Review Board approval and constitute a unified resource for studying disease prediction and clinical reasoning by integrating imaging, structured EHRs data, and clinical documentation.

External dataset

External datasets were assembled for validation only and were processed using the same data pipeline as the internal cohorts to ensure methodological consistency across institutions and modalities. Each external modality mirrors the internal EHRs linkages (radiology reports, clinical notes, laboratory results, CPT procedures, ICD-10 diagnoses, medications) to support comparable evaluation.

- CTPA Dataset: The external CTPA cohort includes 1329 patients evaluated for suspected pulmonary embolism from UCHealth University of Colorado Hospital.
- CCTA Dataset: The external CCTA cohort includes 65 patients assessed for acute or subacute cardiovascular conditions from Second Xiangya Hospital, China.
- Cardiac MRI Dataset: The external cardiac MRI cohort consists of 81 patients undergoing cardiac MRI for suspected or known cardiovascular disease from Second Xiangya Hospital, China.
- RUQ Ultrasound Dataset: The external RUQ ultrasound cohort includes 108 patients undergoing RUQ ultrasound for evaluation of suspected cholecystitis from Stanford Hospitals.

All external datasets were approved by each contributing institution under a data use agreement with Johns Hopkins. We constructed a comprehensive multimodal imaging-EHRs dataset within the Johns Hopkins Health System to evaluate LLM-based clinical reasoning, with and without the proposed clinical preference-aligned retrieval module.

Data splitting and filtering

For the CTPA and Cardiac MRI datasets, we performed patient-level splits of 80% training, 10% validation, and 10% test, using stratification on disease stage, mortality, ICU admission, and readmission status to ensure balanced outcome distributions. The CCTA and ultrasound datasets were split into 80% training and 20% test due to smaller sample sizes; hyperparameters were selected via internal splitting within the training set. All splits were patient-disjoint to prevent data leakage. Test sets were reserved exclusively for final evaluation and assessment of model generalization.

A two-step outlier and preprocessing procedure was applied to all datasets. First, we removed target values lying more than three standard deviations from the mean, recomputed distribution statistics, and then clipped remaining values outside the updated range. This approach reduces noise while retaining biologically plausible extremes. Second, we addressed sparsity in clinical variables. While traditional ML/DL models require explicit imputation, LLMs can interpret missingness when encoded in text form. To provide consistent and interpretable inputs, we applied last-observation-carried-forward (LOCF) imputation⁶⁵ during preprocessing for structured clinical data.

Encoding

Patient trajectories were converted into LLM-readable text using structured templates that transformed EHRs into natural-language sequences, following approaches from Xue et al.⁶⁶ and Liu et al.⁶⁷ (details provided in Supplementary Materials). Each input sequence consisted of five components: (1) patient demographics and history, (2) radiology report, (3) ICD-10 codes with brief explanations, (4) laboratory test results, and (5) a task-specific prompt. Demographic and history fields were included verbatim and did not require imputation for missing values. Output trajectories were encoded using parallel templates that contained only the target variables for the prediction task and were formatted in JSON-style structures to support consistent evaluation. Manually curated templates were used for all input encodings, with standardized output templates described in Supplementary Materials.

LLMs and fine-tuning

We employed the LLaMA-7B biomedical large language model, an openly licensed model built on a widely adopted LLM architecture³³. The model is instruction-tuned and incorporates domain-relevant biomedical knowledge through pretraining on large-scale clinical and scientific text corpora. For Hopkins-LLM, we further fine-tuned LLaMA-7B using a standard cross-entropy objective, applying gradient masking so that updates were restricted to the target output tokens⁶⁸. This approach ensures stable optimization while preserving the integrity of the pretrained language backbone. A full description of fine-tuning hyperparameters and implementation details—as well as comparisons with baseline models—is provided in the Supplementary Materials.

Fine-tuning for diagnostic reasoning

To adapt Hopkins-LLM for clinical diagnostic tasks, we fine-tuned the model on a dataset containing expert-written diagnostic rationales using a *chain-of-thought* (CoT)-based approach⁶⁹. Diagnostic reasoning in clinical practice requires physicians to interpret symptoms, synthesize multimodal information, and articulate intermediate reasoning steps before arriving at a final diagnosis. To capture this multi-step process, we combined CoT fine-tuning with our clinical preference-aligned retrieval module, enabling the model to generate structured rationales that guide downstream predictions and improve diagnostic accuracy.

Formally, given an input prompt

$$\mathbf{I} = w_{1:m}, \quad (1)$$

the language model p_θ generates a response

$$\mathbf{R} = v_{1:n}, \quad (2)$$

representing the CoT rationale followed by the final diagnosis. Training maximizes the conditional likelihood

$$p_\theta(\mathbf{R}|\mathbf{I}) = p_\theta(v_{1:n}|w_{1:m}), \quad (3)$$

using the standard autoregressive cross-entropy loss:

$$\mathcal{L} = \frac{1}{n} \sum_{i=m+1}^{m+n} -\log p_\theta(w_i|w_1, \dots, w_m). \quad (4)$$

Although CoT fine-tuning has been shown to improve reasoning in large language models⁶⁹, acquiring large quantities of high-quality clinical rationales is inherently challenging. Hopkins-LLM therefore incorporates diagnostic priors and a preference-alignment strategy to ensure that generated rationales adhere to established clinical frameworks, improving both interpretability and diagnostic fidelity.

Predictive evaluation

Predictive performance was evaluated using three complementary classification metrics: accuracy, F1 score, and the area under the receiver operating

characteristic curve (AUC-ROC). Accuracy quantified the overall proportion of correctly classified samples and was defined as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (5)$$

The F1 score summarizes the balance between precision and recall, where

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (6)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (7)$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}. \quad (8)$$

To evaluate a model's discrimination across all decision thresholds, we computed the AUC-ROC, defined as the integral of the true positive rate (TPR) over the false positive rate (FPR):

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}), \quad (9)$$

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (10)$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \quad (11)$$

Together, these metrics provide a comprehensive assessment of both threshold-specific and threshold-free predictive performance.

Hyperparameters for Hopkins-LLM fine-tuning

For supervised fine-tuning Hopkins-LLM, we used a learning rate of 1×10^{-5} , a warm-up ratio of 0.1, a batch size of 1 per GPU, and a cosine learning rate scheduler with a weight decay of 0.1, using the AdamW optimizer. Due to memory constraints associated with concatenated radiology reports and structured EHRs inputs, the maximum input sequence length was capped at 3,500–3,800 tokens, depending on the task. Optimal checkpoints were selected based on validation loss, with full multi-task training requiring approximately 18–24 hours on a single NVIDIA A100 80 GB GPU.

Statistical analysis

We report mean values with corresponding standard errors across all experiments. Confidence intervals (CIs) were estimated using a nonparametric bootstrap procedure with 1,000 resamples⁷⁰. Additional performance metrics—including precision, recall, and ROC-AUC—were computed using both macro-averaging (unweighted) and micro-averaging (sample-weighted) formulations. ROC-AUC values were also evaluated using sample-consensus (SC) agreement frequency⁷¹. All statistical tests were two-sided, with a significance threshold of $P < 0.05$. Differences between model performances were assessed using two-sided paired t-tests.

PESI

The Pulmonary Embolism Severity Index (PESI)³⁷ is a validated clinical risk score used to stratify patients with suspected pulmonary embolism. It incorporates demographic, comorbidity, and physiological variables to estimate short-term mortality risk and guide management decisions. For Hopkins-LLM, we extracted PESI-related variables as part of the clinical preference-aligned retrieval module, including:

- Age
- Sex
- History of cancer

- History of heart failure
- History of chronic lung disease
- Heart rate ≥ 110
- Systolic BP < 100 mmHg
- Respiratory rate ≥ 30
- Temperature $< 36^\circ\text{C}/96.8^\circ\text{F}$
- Altered mental status (disorientation, lethargy, stupor, or coma)
- O₂ saturation $< 90\%$

Only PESI-relevant variables were parsed for LLM inference in the PE cohort.

ASCVD

The Atherosclerotic Cardiovascular Disease (ASCVD) 10-year risk estimator³⁸ predicts the likelihood of major cardiovascular events such as myocardial infarction or stroke. Variables used in ASCVD scoring and included in our alignment module are:

- Age
- Sex
- Race
- Systolic Blood Pressure (mm Hg)
- Diastolic Blood Pressure (mm Hg)
- Total Cholesterol (mg/dL)
- HDL Cholesterol (mg/dL)
- LDL Cholesterol (mg/dL)
- History of Diabetes?
- Smoker?
- On Hypertension Treatment?
- On a Statin?
- On Aspirin Therapy?

CONFIRM

The CONFIRM score (Coronary Artery Disease Evaluation and Management)³⁶ evaluates the severity of coronary artery disease using a combination of demographic, clinical, and imaging-based risk factors. The CONFIRM module incorporates the following variables:

- Age
- Sex
- Race
- Diabetes Mellitus
- Hypertension
- Dyslipidemia
- Current or Previous Smoking
- Left Ventricular Function (EF%) (ejection fraction, indicating heart function)
- Coronary Calcium Score

Tokyo Guidelines (TG13/TG18) for Acute Cholecystitis

The Tokyo Guidelines⁷² provide standardized criteria for diagnosing acute cholecystitis, integrating clinical symptoms, laboratory abnormalities, and ultrasound findings. Variables mapped to ICD-10 codes and used for ultrasound-based alignment included:

- Murphy's Sign: R10.819 (Right upper quadrant abdominal tenderness)
- Right upper abdominal quadrant mass, pain, or tenderness: R10.811, R10.819, R19.00 (Right upper quadrant pain)
- Fever: R50.9 (Fever, unspecified)
- Elevated CRP (C-reactive protein): R79.82 (Elevated C-reactive protein)
- Elevated WBC count: D72.829 (Elevated white blood cell count, unspecified)
- Acute cholecystitis with gallstones: K81.0
- Acute cholecystitis without gallstones: K81.9

- Pericholecystic fluid (often associated with gallbladder inflammation): Covered under K81.0
- Gallstones (Cholelithiasis) without cholecystitis: K80.20

Data availability

The multimodal imaging and EHRs datasets used in this study were derived from the Johns Hopkins Health System under Institutional Review Board (IRB) approval and contain protected health information. Due to patient privacy considerations, the raw data cannot be publicly shared. De-identified subsets of the data may be made available upon reasonable request to the corresponding author, subject to completion of a Data Use Agreement (DUA) and approval by the Johns Hopkins IRB.

Code availability

All deep learning models were implemented in Python (version 3.10) using PyTorch (version 2.1.2). The following libraries were used for model development and evaluation: NumPy (1.26.4), pandas (2.2.1), transformers (4.36.1), vLLM (0.2.5), scikit-learn (1.2.1), matplotlib (3.7.1), and SciPy (1.11.3). Custom code modules were used for data input/output pipelines and distributed parallelization across computing nodes and GPUs. All source code supporting this study is available for scientific research and non-commercial use at: https://github.com/YuliWanghust/Hopkins_LLM.

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Competing interests

Harrison Bai, MD, serves as an Associate Editor of the npj Digital Medicine. He was not involved in the peer-review process or editorial decision-making for this manuscript.

Additional information

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